

Smart AI Planner

Team 59

Senior Design Team Contract

University of Cincinnati

College of Education, Criminal Justice and Human Services

School of Information Technology

Bonnie Hou

Fakhruddin Shaik

Callie Jacobs

Bilal Ahmadzai

Table of Contents

<i>Intent</i>	<i>3</i>
<i>Senior Design Contract.....</i>	<i>4</i>
Project Summary	4
Problem Statement.....	4
Solution	4
Contact Information.....	4
Sponsor Information (if applicable)	4
Project Source.....	4
Project Objectives/Goals	4
Team Members and Responsibilities	5
Project Scope	5
Quick Project Timeline	5
Technologies Used	5
Ethical Considerations	5
Team Rules	5
<i>Team Signatures:.....</i>	<i>6</i>
<i>References.....</i>	<i>7</i>

Intent

The following contract was written and agreed upon by ***Bonnie Hou, Fakhruddin Shaik, Callie Jacobs, and Bilal Ahmadzai***. The contract provides expectations, objectives, and results for developing the **Smart AI Planner**.

The contract is effective for all team members participating in the Senior Design Capstone class series in the 2025-2026 academic year.

Senior Design Contract

Project Summary

The Smart planner app is an AI powered solution designed to help students overcome the problems of poor planning and time management. By integrating academic calendars, job schedules, and extracurricular activities into one platform, the app provides dynamic scheduling and smart prioritization.

Problem Statement

Managing time effectively has become a significant challenge for college students. Students are constantly juggling multiple assignments, classes, jobs, and other extracurricular activities. Without structured planning, students are likely to experience academic difficulties and mental health challenges. According to *Journal and Further Higher Education*, research highlights that poor task planning, insufficient progress monitoring, emotional depression, high levels of neuroticism, low conscientiousness, and poorer academic performance all contribute to procrastination and psychological distress in higher education (Faure-Carvallo, Nieto-Fernández, Calderon, & Gustems, 2025). Traditional planners and calendars are becoming ineffective because they require manual input and fail to adapt to personal habits. Therefore, students need more intelligent and adaptive solutions.

Solution

To address these challenges, a Smart Planner App powered by AI is essential. This app changes traditional planning and is developed into dynamic and adaptative systems that respond to students' daily needs and habits.

Key features of Smart Planner:

- Auto syncing with academic calendars: Automatically imports class schedules, deadlines, and exams, reducing manual input and ensuring accuracy.
- Integration with jobs and extracurricular activities: Connects with work and extracurricular activities so students can balance academic and non-academic responsibilities.
- Dynamic scheduling and rescheduling: Adjusts plans in real time and proposes new schedules when tasks or events are missed.
- Smart prioritization of tasks: Rank tasks based on urgency and importance to minimize procrastination.
- Smart reminders and alerts: Provides notifications that anticipate upcoming tasks and deadlines.
- Progress monitoring and insights: Tracks progress and productivity, offering feedback to improve habits.

Contact Information

TEAM MEMBER	DEGREE + TRACK TRACK N/A FOR BSCYBER	EMAIL	PHONE NUMBER OR OTHER CONTACT INFO
Bonnie Hou	BSIT-Cyber	hoube@mail.uc.edu	513-806-9775
Fakhruddin Shaik	BSIT – Data Tech	shaikfn@mail.uc.edu	513-996-8082
Callie Jacobs	BSIT-Cybersecurity	Jacobsck@mail.uc.edu	513-467-8979
Bilal Ahmadzai	Network Administration	Ahmadzba@ucmail.uc.edu	513-915-1061

Project Source

This project idea originated from the team’s shared experiences balancing academic, work, and personal obligations. The concept was conceived after recognizing the limitations of traditional planners and the increasing demand for AI-driven organization tools among students. The team collaboratively conducted requirements analysis through:

- Reviewing academic studies on student time-management challenges
- Interviewing peers to identify common scheduling points
- Studying existing planning apps to evaluate gaps in current solutions

The team formed during the Senior Design course selection process based on a mutual desire to develop an impactful productivity tool.

Project Objectives/Goals

The goal of the Smart Planner App is to help students reduce stress, minimize procrastination, and improve academic outcomes by providing a responsive and intelligent planning system. Smart Planner App aims to provide an intelligent system that reduces procrastination, improves organization, and supports academic success.

High Level Goals and Expected Impact:

- Lower Stress
- Reduce Procrastination
- Enhance academic performance
- Improve productivity habits

Team Members and Responsibilities

Team Member	Role	Responsibilities
-------------	------	------------------

Callie Jacobs	Backend Developer	APIs, server-side logic, scheduling engine, database operations, calendar synchronization, deployment, and testing.
Bonnie Hou	Cybersecurity Specialist	Data privacy, security controls, secure authentication, compliance considerations
Fakhruddin Shaik	Frontend and Data Integration Lead	UI/UX, dashboard, calendar/task screens, frontend integration, and accessibility.
Bilal Ahmadzai	Network & DevOps Engineer	Data Visualization, technical documentation, Network configuration, server and cloud infrastructure, deployment support, troubleshooting, and ensuring reliable communication between frontend, backend, and external services.

Project Scope

Our team will develop a functional application that enables student users to improve time management using AI-powered scheduling. The project scope includes:

In-Scope Features

- User registration and secure login
- Calendar syncing (academic + work) - from and to system
- AI task prioritization
- Intelligent reminders - based off prioritization
- Dynamic event creation/schedule adjustments
- Dashboard with productivity insights

Out-of-Scope (for Capstone timeline)

- Full mobile app deployment to app stores
- Advanced machine-learning recommendation engine
- Integration with external APIs beyond school/work calendars
- Voice assistant features

The project will be structured into milestones so it can be completed within two semesters of the Senior Design timeline.

*“Our team will develop a functional application that enables users to solve users to **solve time-management and scheduling challenges** by utilizing the following features and functionality.”*

Quick Project Timeline – 23 weeks excluding holidays (tentative past task 7)

<i>Task #</i>	<i>Task Name</i>	<i>Duration</i>	<i>Start Date</i>	<i>End Date</i>
1	Base Database/Calendar Creation	2 weeks	Sept. 2, 2026	Oct 5, 2026
2	Import schedule information	1 weeks	Oct. 5, 2026	Oct. 12, 2026
3	Export schedule information	2 weeks	Oct. 12, 2026	Oct. 26, 2026
4	Website (sans features)	3 weeks	Oct. 5, 2026	Oct. 26, 2026
5	Visual Integration and schedule work	2 weeks	Oct. 26, 2026	Nov. 9, 2026
6	Dynamic scheduling and rescheduling	4 weeks	Nov. 9, 2026	Dec. 13, 2026
7	Task Prioritization	3 weeks	Jan. 2027	Jan. 2027
8	Smart Reminders and alerts	4 weeks	Feb. 2027	Feb. 2027
9	Progress monitoring and insights	3 weeks	Mar. 2027	Mar. 2027
10	User Registration and secure login	2 weeks	Mar. 2027	Mar. 2027

Technologies Used

- **Frontend:** React Native or Flutter
- **Backend:** Node.js or Python
- **Database:** Firebase, PostgreSQL, or MongoDB
- **AI/ML:** TensorFlow Lite or custom Python models
- **Authentication:** OAuth 2.0 / Firebase Auth
- **Hosting:** AWS, Google Cloud Platform or Azure

Ethical Considerations

Building an AI-driven planner requires careful attention to user data privacy, transparency, and responsible usage of AI. Students entrust the system with sensitive schedule data, personal habits, and academic workloads. Research stresses the importance of implementing secure

data practices and minimizing psychological pressure created by productivity systems (Faure-Carvallo et al., 2025).

Ethical considerations include:

- Ensuring secure storage of user information
- Avoiding manipulative or overly intrusive AI nudges
- Maintaining transparency in how AI makes scheduling decisions
- Complying with FERPA-related considerations for academic data

Team Rules

1. *Plagiarism will not be tolerated. Any team member that plagiarizes will be subject to university policies, and a team meeting will be called.*
2. *Each team member will stay current on their tasks to ensure the project milestones are met. If an event conflicts that will affect the completion of a deliverable, the team member will notify the other team members at least 24 hours in advance of the scheduled due date.*
3. *If a group member will be absent on class days or for an extended period, they will notify the other team members and the instructors.*
4. All team members must communicate openly and respectfully.
5. All decisions should be made collaboratively, with documentation of major changes.

Team Signatures:

Signature: Bonnie Hou

Date: 9/19/2026

Signature: Fakhruddin Shaik

Date: 9/19/2026

Signature: Bilal Ahmadzai

Date: 9/19/2026

Signature: Callie Jacobs

Date: 9/19/2026



References

- Faure-Carvallo, A., Nieto-Fernández, S., Calderon, C., & Gustems, J. (2025). *Relationship between procrastination, time management, personality, and psychological distress in higher education*. Journal of Further and Higher Education, 49(4), 417–429. <https://doi.org/10.1080/0309877X.2025.2459852>
- Hoon Yi Xuen, A., Ngai Zhi Xuan, N., & Mariappan, L. (2023). *Investigating the factors that affect students to manage their time properly*. International Journal of Research Publication and Reviews, 4(10), 267–276. <https://ijrpr.com/uploads/V4ISSUE10/IJRPR18045.pdf>